

Cumulative Final Review: A Geometric Decision Framework

Thursday, December 10, 2026

Visualization: State space, diagnostic data, and global method		
system	diagnostic data	global method
$\dot{x} = f(x)$	$f(x_*) = 0$, sign f	phase line and uniqueness
$\dot{\theta} = f(\theta)$ on S^1	$f(\theta_*) = 0$, sign f	$f \neq 0$: $T = \left \int_0^{2\pi} \frac{d\theta}{f(\theta)} \right $
$\dot{z} = Az$ $\dot{z} = F(z)$ in $\mathbb{R}^2$	$\text{tr } A$, $\det A$, $(\text{tr } A)^2 - 4 \det A$ $F(z_*) = 0$, $DF(z_*)$	eigendirections and exact linear flow invariance, energy, symmetry, global theorems

Observation: Hypothesis for the circle-period formula

The circle-period formula applies when $f(\theta) \neq 0$ for every $\theta \in S^1$; continuity then forces f to have one sign on S^1 .

Observation: Scope of nonlinear linearization

For C^1 nonlinear planar systems, a hyperbolic Jacobian gives only the local portrait near the equilibrium.

Recall: Local conclusion

A **local conclusion** holds on a neighborhood of a specified state.

Recall: Global conclusion

A **global conclusion** holds on an explicitly stated entire domain.

Recall: Necessary condition

A condition is **necessary** for a conclusion if the conclusion implies the condition.

Recall: Sufficient condition

A condition is **sufficient** for a conclusion if the condition implies the conclusion.

Question: A scalar saddle-node

Analyze

$$\dot{x} = \lambda + x^2$$

for every real λ .

Solution. If $\lambda < 0$, the equilibria are

$$x_- = -\sqrt{-\lambda}, \quad x_+ = \sqrt{-\lambda}.$$

Since $f'(x) = 2x$, the equilibrium x_- is asymptotically stable and x_+ is unstable. The phase line is

$$\begin{array}{c|cccccc} x & (-\infty, x_-) & x_- & (x_-, x_+) & x_+ & (x_+, \infty) \\ \hline \dot{x} & + & 0 & - & 0 & + \end{array}$$

At $\lambda = 0$, the origin attracts from the left and repels from the right:

$$\begin{array}{c|ccc} x & (-\infty, 0) & 0 & (0, \infty) \\ \hline \dot{x} & + & 0 & + \end{array}$$

Thus the origin is half-stable and Lyapunov unstable. If $\lambda > 0$, there is no equilibrium and $\dot{x} > 0$ everywhere.

Question: Saddle-node bottleneck time

Let $A > 0$ and $\lambda > 0$. For

$$\dot{x} = \lambda + x^2,$$

compute the travel time from $x = -A$ to $x = A$ and its leading behavior as $\lambda \downarrow 0$.

Solution. The travel time is

$$\begin{aligned} T_A(\lambda) &= \int_{-A}^A \frac{dx}{\lambda + x^2} \\ &= \frac{2}{\sqrt{\lambda}} \arctan\left(\frac{A}{\sqrt{\lambda}}\right) \sim \frac{\pi}{\sqrt{\lambda}} \quad (\lambda \downarrow 0). \end{aligned}$$

Question: The saddle-node on a circle

Let $a > 0$. Analyze

$$\dot{\theta} = \omega - a \sin \theta, \quad \theta \in S^1,$$

for every $\omega \in \mathbb{R}$.

Solution. If $|\omega| < a$, define

$$\alpha = \arcsin(\omega/a), \quad -\frac{\pi}{2} < \alpha < \frac{\pi}{2}.$$

The equilibria modulo 2π are

$$\theta_s = \alpha, \quad \theta_u = \pi - \alpha.$$

Since

$$f'(\theta) = -a \cos \theta,$$

θ_s is asymptotically stable and θ_u is unstable. At $|\omega| = a$, they coalesce in a saddle-node. At $\omega = a$, the fixed point $\theta = \pi/2$ attracts from the left and repels from the right. At $\omega = -a$, the fixed point $\theta = 3\pi/2$ attracts from the right and repels from the left. In either threshold case the fixed point is half-stable and Lyapunov unstable. If $|\omega| > a$, there is no equilibrium and the phase rotates with the sign of ω .

Question: Rotation period beyond the circle saddle-node

Let $a > 0$ and $|\omega| > a$. Compute the positive rotation period of

$$\dot{\theta} = \omega - a \sin \theta, \quad \theta \in S^1.$$

Solution. For $\omega > a$, set $\phi = \theta - \pi/2$ and $u = \tan(\phi/2)$. Then

$$\begin{aligned} T &= \int_0^{2\pi} \frac{d\theta}{\omega - a \sin \theta} \\ &= \int_{-\pi/2}^{3\pi/2} \frac{d\phi}{\omega - a \cos \phi} \\ &= \int_{-\pi}^{\pi} \frac{d\phi}{\omega - a \cos \phi} \quad \text{by } 2\pi\text{-periodicity} \\ &= \int_{-\infty}^{\infty} \frac{2 du}{(\omega - a) + (\omega + a)u^2} \\ &= \frac{2\pi}{\sqrt{\omega^2 - a^2}}. \end{aligned}$$

For $\omega < -a$, the orientation reverses and the positive period has the same formula.

Recall: Trace–determinant notation

For a real 2×2 matrix A , write

$$\tau = \operatorname{tr} A, \quad \Delta = \det A, \quad D = \tau^2 - 4\Delta.$$

Visualization: Trace–determinant classification

For the linear system $\dot{z} = Az$,

condition	linear type
$\Delta < 0$	saddle
$\Delta > 0, D > 0$	node; stability from sign τ
$\Delta > 0, D < 0, \tau \neq 0$	spiral; stability from sign τ
$\Delta > 0, D < 0, \tau = 0$	center
$D = 0, \Delta > 0$	star or degenerate node; check eigenvectors
$\Delta = 0$	non-isolated equilibrium set

Negative trace means asymptotically stable; positive trace means unstable whenever $\Delta > 0$ and the equilibrium is hyperbolic.

Question: The same Jacobian can hide different nonlinear behavior

Compare

$$\dot{x} = y, \quad \dot{y} = -x$$

with

$$\dot{x} = y, \quad \dot{y} = -x - y(x^2 + y^2).$$

Solution. Both systems have

$$DF(0,0) = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

and eigenvalues $\pm i$. For the first system,

$$E(x,y) = \frac{1}{2}(x^2 + y^2)$$

satisfies $\dot{E} = 0$, so its positive level sets are periodic orbits and the origin is a center.

For the second system,

$$\dot{E} = x\dot{x} + y\dot{y} = -y^2(x^2 + y^2) \leq 0.$$

Thus each solution stays in the closed disk determined by its initial energy and exists for all $t \geq 0$. Since its energy is nonincreasing and nonnegative, write

$$E(x(t), y(t)) \rightarrow L \geq 0.$$

Its omega-limit set is nonempty and invariant, and every point in it has energy L . If $L > 0$, energy must stay constant along the trajectories in this set, so $\dot{E} = -2Ly^2 = 0$ forces $y = 0$. But on $y = 0$ the equation $\dot{y} = -x$ prevents any nonzero point from staying on that line. This contradicts invariance. Therefore $L = 0$, and

$$(x(t), y(t)) \rightarrow (0, 0) \quad (t \rightarrow \infty).$$

Question: The same Jacobian can hide different nonlinear behavior (continued)

Also,

$$\|(x(t), y(t))\| \leq \|(x(0), y(0))\|$$

proves Lyapunov stability. Thus the origin of the nonlinear system is globally asymptotically stable. Every solution converges to it, so there is no nonconstant periodic orbit.

Question: A parameter creates a unique stable limit cycle

Let $\mu \in \mathbb{R}$. Analyze

$$\dot{x} = \mu x - y - (x^2 + y^2)x,$$

$$\dot{y} = x + \mu y - (x^2 + y^2)y.$$

Solution. At the origin,

$$DF(0,0) = \begin{pmatrix} \mu & -1 \\ 1 & \mu \end{pmatrix}, \quad \lambda = \mu \pm i.$$

For $r > 0$,

$$\begin{aligned} r\dot{r} &= x\dot{x} + y\dot{y} \\ &= \mu r^2 - r^4, \end{aligned}$$

so

$$\dot{r} = r(\mu - r^2).$$

Also,

$$\dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} = 1.$$

Therefore

$$\begin{array}{l|l} \mu < 0 & \dot{r} < 0 \text{ for } r > 0, \quad r(t) \rightarrow 0 \\ \mu = 0 & \dot{r} = -r^3 < 0 \text{ for } r > 0, \quad r(t) \rightarrow 0 \\ \mu > 0 & \dot{r} > 0 \text{ for } 0 < r < \sqrt{\mu}, \quad \dot{r} < 0 \text{ for } r > \sqrt{\mu} \end{array}$$

Thus the origin is globally asymptotically stable for $\mu \leq 0$ and unstable for $\mu > 0$. When $\mu > 0$, the invariant circle $r = \sqrt{\mu}$ attracts every solution with $r(0) > 0$ and has period 2π . On every other positive-radius interval, $r(t)$ is strictly monotone, so no other periodic orbit exists. Thus $r = \sqrt{\mu}$ is the unique stable limit cycle.

Visualization: Criteria for periodic-orbit questions

criterion	conclusion
$\nabla \cdot (BF) < 0$ away from equilibria	no periodic orbit
$\nabla \cdot (BF)$ has one strict sign on a simply connected domain	no periodic orbit in that domain
compact trapping region with no equilibrium	at least one periodic orbit
isolated closed orbit	limit cycle
all Liénard hypotheses	exactly one stable limit cycle

Question: A scaled Liénard oscillator

Let $a > 0$ and $\mu > 0$. Prove that

$$\ddot{x} + \mu(x^2 - a^2)\dot{x} + x = 0$$

has exactly one stable limit cycle.

Solution. For Liénard's theorem,

$$f(x) = \mu(x^2 - a^2), \quad g(x) = x,$$

and

$$F(x) = \int_0^x f(u) du = \mu \left(\frac{x^3}{3} - a^2 x \right).$$

The function g is odd and positive for $x > 0$, and f is even. The only positive zero of F is

$$x = \sqrt{3}a.$$

Moreover,

$$F(x) < 0 \quad (0 < x < \sqrt{3}a),$$

and for $x > \sqrt{3}a$,

$$F(x) > 0, \quad F'(x) = \mu(x^2 - a^2) > 0, \quad F(x) \rightarrow \infty \quad (x \rightarrow \infty).$$

Thus there is exactly one stable limit cycle for every $\mu > 0$.

Question: Period of the scaled relaxation oscillator

Fix $a > 0$. Estimate the period of the stable limit cycle of

$$\ddot{x} + \mu(x^2 - a^2)\dot{x} + x = 0$$

as $\mu \rightarrow \infty$.

Solution. Set $x = aX$ and divide the equation by a :

$$\ddot{X} + \mu a^2(X^2 - 1)\dot{X} + X = 0.$$

This is the van der Pol equation with parameter $\nu = \mu a^2$. The scaling changes amplitude but leaves time, and hence period, unchanged. Since $a > 0$ is fixed, $\mu \rightarrow \infty$ implies $\nu \rightarrow \infty$. The van der Pol period estimate therefore gives

$$T(\mu) = \nu(3 - 2 \ln 2) + o(\nu).$$

Substituting $\nu = \mu a^2$ gives

$$\boxed{T(\mu) = \mu a^2(3 - 2 \ln 2) + o(\mu)} \quad (\mu \rightarrow \infty).$$

Question: Audit a planar omega-limit conclusion

For a C^1 planar vector field, a positive semi-orbit lies in a compact subset of the vector field's domain, and its omega-limit set contains no equilibrium. What follows?

Answer. The Poincaré–Bendixson theorem makes the omega-limit set a periodic orbit.

Question: Audit an annulus conclusion

A compact annulus contains no equilibrium. Does a periodic orbit follow?

Answer. No. Positive invariance, or another form of forward confinement, must also be proved.

Question: Audit a first-integral conclusion

For a C^1 planar system, a connected compact regular level component of a first integral contains no equilibrium. What follows?

Answer. The level component is a periodic orbit. Nearby energy levels may form a family, so the orbit need not be a limit cycle.

Question: Audit a divergence conclusion

The divergence of a C^1 planar vector field is negative on an open, simply connected region containing a proposed closed orbit. What follows?

Answer. The Bendixson–Dulac criterion with $B \equiv 1$ rules out the proposed closed orbit.

Idea: Four layers of a complete qualitative analysis

state space and parameters $\Rightarrow$ equilibria and local data
 $\Rightarrow$ global invariant structure $\Rightarrow$ exact conclusion.